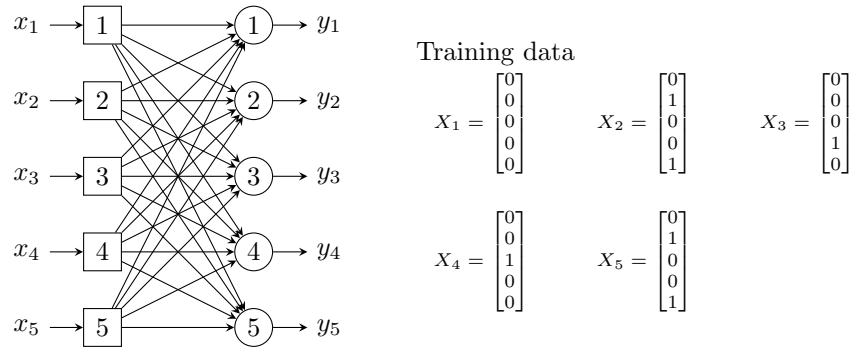


Further explanation on Hebbian network calculation

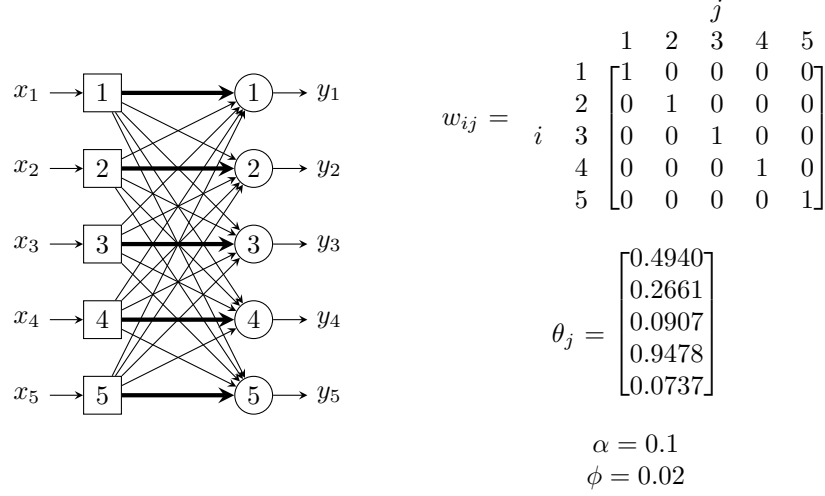
Richard W.T.K.



This means there are five (5) sets of training data. In first set of training data X_1 , $x_1 = 0$, $x_2 = 0$, $x_3 = 0$, $x_4 = 0$, and $x_5 = 0$; in second set of training data X_2 , $x_1 = 0$, $x_2 = 1$, $x_3 = 0$, $x_4 = 0$, and $x_5 = 1$. Alternatively the training data can be presented in a table form as follows.

Data set no.	x_1	x_2	x_3	x_4	x_5
1 (X_1)	0	0	0	0	0
2 (X_2)	0	1	0	0	1
3 (X_3)	0	0	0	1	0
4 (X_4)	0	0	1	0	0
5 (X_5)	0	1	0	0	1

1 Initialisation



The weights are presented in matrix form. With the notation of w_{ij} , i is the input neuron number and the j is the output neuron number (w_{ij} is the weight going from neuron i to neuron j). From the weight matrix, it is seen that only w_{11} , w_{22} , w_{33} , w_{44} , and w_{55} are initialised to have the value of 1 and the rest of the weights have the value of 0.

Thresholds are also presented in matrix form as shown in θ_j . As input neurons have no threshold, only the five output neurons have threshold. Therefore the threshold matrix has only five values for θ_1 , θ_2 , θ_3 , θ_4 , and θ_5 . Threshold for output neuron 1, θ_1 is 0.4940, threshold for output neuron 2, θ_2 is 0.2661, and so on.

The learning rate, α is set as 0.1, and the forgetting factor, ϕ is set as 0.02.

2 Iteration 1: Activation

In the first iteration the first set of training data, X_1 is used to train the network. With X_1 , we have

$$\begin{aligned} x_1 &= 0 \\ x_2 &= 0 \\ x_3 &= 0 \\ x_4 &= 0 \\ x_5 &= 0 \end{aligned} \quad \text{or} \quad X_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

To activate the neurons, we calculate the output of the neurons with the formula

$$y_j = \text{step} \left(\sum_{i=1}^n x_i w_{ij} - \theta_j \right)$$

For the output of first neuron, y_1 , it is calculated with

$$\begin{aligned}
y_1 &= \text{step} \left(\sum_{i=1}^5 x_i w_{i1} - \theta_1 \right) \\
&= \text{step} (x_1 w_{11} + x_2 w_{21} + x_3 w_{31} + x_4 w_{41} + x_5 w_{51} - \theta_1) \\
&= \text{step} (0(1) + 0(0) + 0(0) + 0(0) + 0(0) - 0.4940) = \text{step} (-0.4940) = 0
\end{aligned}$$

For the output of second neuron, y_2 , it is calculated with

$$\begin{aligned}
y_2 &= \text{step} \left(\sum_{i=1}^5 x_i w_{i2} - \theta_2 \right) \\
&= \text{step} (x_1 w_{12} + x_2 w_{22} + x_3 w_{32} + x_4 w_{42} + x_5 w_{52} - \theta_2) \\
&= \text{step} (0(0) + 0(1) + 0(0) + 0(0) + 0(0) - 0.2661) = \text{step} (-0.2661) = 0
\end{aligned}$$

For the output of third neuron, y_3 , it is calculated with

$$\begin{aligned}
y_3 &= \text{step} \left(\sum_{i=1}^5 x_i w_{i3} - \theta_3 \right) \\
&= \text{step} (x_1 w_{13} + x_2 w_{23} + x_3 w_{33} + x_4 w_{43} + x_5 w_{53} - \theta_3) \\
&= \text{step} (0(0) + 0(0) + 0(1) + 0(0) + 0(0) - 0.0907) = \text{step} (-0.0907) = 0
\end{aligned}$$

For the output of fourth neuron, y_4 , it is calculated with

$$\begin{aligned}
y_4 &= \text{step} \left(\sum_{i=1}^5 x_i w_{i4} - \theta_4 \right) \\
&= \text{step} (x_1 w_{14} + x_2 w_{24} + x_3 w_{34} + x_4 w_{44} + x_5 w_{54} - \theta_4) \\
&= \text{step} (0(0) + 0(0) + 0(0) + 0(1) + 0(0) - 0.9478) = \text{step} (-0.9478) = 0
\end{aligned}$$

For the output of fifth neuron, y_5 , it is calculated with

$$\begin{aligned}
y_5 &= \text{step} \left(\sum_{i=1}^5 x_i w_{i5} - \theta_5 \right) \\
&= \text{step} (x_1 w_{15} + x_2 w_{25} + x_3 w_{35} + x_4 w_{45} + x_5 w_{55} - \theta_5) \\
&= \text{step} (0(0) + 0(0) + 0(0) + 0(0) + 0(5) - 0.0737) = \text{step} (-0.0737) = 0
\end{aligned}$$

These calculations of the value of the output neurons can be presented as a matrix operation.

$$Y = \text{step} (W^T X - \theta)$$

W is the weight matrix, and W^T is the transpose of the weight matrix, meaning the row and columns are switched.

$$W = \begin{bmatrix} w_{11} & w_{12} & w_{13} & w_{14} & w_{15} \\ w_{21} & w_{22} & w_{23} & w_{24} & w_{25} \\ w_{31} & w_{32} & w_{33} & w_{34} & w_{35} \\ w_{41} & w_{42} & w_{43} & w_{44} & w_{45} \\ w_{51} & w_{52} & w_{53} & w_{54} & w_{55} \end{bmatrix}$$

$$W^T = \begin{bmatrix} w_{11} & w_{12} & w_{13} & w_{14} & w_{15} \\ w_{21} & w_{22} & w_{23} & w_{24} & w_{25} \\ w_{31} & w_{32} & w_{33} & w_{34} & w_{35} \\ w_{41} & w_{42} & w_{43} & w_{44} & w_{45} \\ w_{51} & w_{52} & w_{53} & w_{54} & w_{55} \end{bmatrix}$$

Substituting the values into the formula,

$$\begin{aligned} Y &= \text{step} (W^T X - \theta) \\ &= \text{step} \left(\begin{bmatrix} w_{11} & w_{12} & w_{13} & w_{14} & w_{15} \\ w_{21} & w_{22} & w_{23} & w_{24} & w_{25} \\ w_{31} & w_{32} & w_{33} & w_{34} & w_{35} \\ w_{41} & w_{42} & w_{43} & w_{44} & w_{45} \\ w_{51} & w_{52} & w_{53} & w_{54} & w_{55} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} - \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \\ \theta_5 \end{bmatrix} \right) \end{aligned}$$

The multiplication of two matrices is performed by summing the multiplication of the elements in m row of the first matrix with the n column of the second matrix to get the value of the element of row m column n of the resultant matrix. If you have not learnt matrix operation before this might be new for you.

$$\begin{aligned} Y &= \text{step} \left(\begin{bmatrix} w_{11}x_1 + w_{12}x_2 + w_{13}x_3 + w_{14}x_4 + w_{15}x_5 \\ w_{21}x_1 + w_{22}x_2 + w_{23}x_3 + w_{24}x_4 + w_{25}x_5 \\ w_{31}x_1 + w_{32}x_2 + w_{33}x_3 + w_{34}x_4 + w_{35}x_5 \\ w_{41}x_1 + w_{42}x_2 + w_{43}x_3 + w_{44}x_4 + w_{45}x_5 \\ w_{51}x_1 + w_{52}x_2 + w_{53}x_3 + w_{54}x_4 + w_{55}x_5 \end{bmatrix} - \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \\ \theta_5 \end{bmatrix} \right) \\ &= \text{step} \left(\begin{bmatrix} w_{11}x_1 + w_{12}x_2 + w_{13}x_3 + w_{14}x_4 + w_{15}x_5 - \theta_1 \\ w_{21}x_1 + w_{22}x_2 + w_{23}x_3 + w_{24}x_4 + w_{25}x_5 - \theta_2 \\ w_{31}x_1 + w_{32}x_2 + w_{33}x_3 + w_{34}x_4 + w_{35}x_5 - \theta_3 \\ w_{41}x_1 + w_{42}x_2 + w_{43}x_3 + w_{44}x_4 + w_{45}x_5 - \theta_4 \\ w_{51}x_1 + w_{52}x_2 + w_{53}x_3 + w_{54}x_4 + w_{55}x_5 - \theta_5 \end{bmatrix} \right) \\ &= \text{step} \left(\begin{bmatrix} 1(0) + 0(0) + 0(0) + 0(0) + 0(0) - 0.4940 \\ 0(0) + 1(0) + 0(0) + 0(0) + 0(0) - 0.2661 \\ 0(0) + 0(0) + 1(0) + 0(0) + 0(0) - 0.0907 \\ 0(0) + 0(0) + 0(0) + 1(0) + 0(0) - 0.9478 \\ 0(0) + 0(0) + 0(0) + 0(0) + 1(0) - 0.0737 \end{bmatrix} \right) \end{aligned}$$

$$= \text{step} \left(\begin{bmatrix} -0.4940 \\ -0.2661 \\ -0.0907 \\ -0.9478 \\ -0.0737 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix}$$

This is equivalent to the previous calculation shown for y_1, y_2, y_3, y_4 , and y_5 .

3 Iteration 1: Learning

The weights are then corrected with the generalised activity product rule

$$\Delta w_{ij} = \phi y_j (\lambda x_i - w_{ij})$$

From the activation, all output neurons have the value of zero (0). Therefore from the weight correction formula, the weight correction Δw_{ij} will be 0 for all weights. With zero weight correction, the new weights will remain the same.

$$w_{ij} = \begin{matrix} & & & j & & \\ & & & 1 & 2 & 3 & 4 & 5 \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & i & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

4 Iteration 2: Activation

The second set set of training data, X_2 is used to train the network in the second iteration. With X_2 , we have

$$\begin{matrix} x_1 = 0 \\ x_2 = 1 \\ x_3 = 0 \\ x_4 = 0 \\ x_5 = 1 \end{matrix} \quad \text{or} \quad X_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{aligned} Y &= \text{step} (W^T X - \theta) \\ &= \text{step} \left(\begin{bmatrix} w_{11} & w_{21} & w_{31} & w_{41} & w_{51} \\ w_{12} & w_{22} & w_{32} & w_{42} & w_{52} \\ w_{13} & w_{23} & w_{33} & w_{43} & w_{53} \\ w_{14} & w_{24} & w_{34} & w_{44} & w_{54} \\ w_{15} & w_{25} & w_{35} & w_{45} & w_{55} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} - \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \\ \theta_5 \end{bmatrix} \right) \end{aligned}$$

$$\begin{aligned}
&= \text{step} \left(\begin{bmatrix} w_{11}x_1 + w_{21}x_2 + w_{31}x_3 + w_{41}x_4 + w_{51}x_5 - \theta_1 \\ w_{12}x_1 + w_{22}x_2 + w_{32}x_3 + w_{42}x_4 + w_{52}x_5 - \theta_2 \\ w_{13}x_1 + w_{23}x_2 + w_{33}x_3 + w_{43}x_4 + w_{53}x_5 - \theta_3 \\ w_{14}x_1 + w_{24}x_2 + w_{34}x_3 + w_{44}x_4 + w_{54}x_5 - \theta_4 \\ w_{15}x_1 + w_{25}x_2 + w_{35}x_3 + w_{45}x_4 + w_{55}x_5 - \theta_5 \end{bmatrix} \right) \\
&= \text{step} \left(\begin{bmatrix} 1(0) + 0(1) + 0(0) + 0(0) + 0(1) - 0.4940 \\ 0(0) + 1(1) + 0(0) + 0(0) + 0(1) - 0.2661 \\ 0(0) + 0(1) + 1(0) + 0(0) + 0(1) - 0.0907 \\ 0(0) + 0(1) + 0(0) + 1(0) + 0(1) - 0.9478 \\ 0(0) + 0(1) + 0(0) + 0(0) + 1(1) - 0.0737 \end{bmatrix} \right) \\
&= \text{step} \left(\begin{bmatrix} -0.4940 \\ 0.7339 \\ -0.0907 \\ -0.9478 \\ 0.9263 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix}
\end{aligned}$$

5 Iteration 2: Learning

The weights are then corrected with the generalised activity product rule

$$\Delta w_{ij} = \phi y_j (\lambda x_i - w_{ij})$$

From the formula, if the output neuron has the value of 0, the weight correction will be 0. Therefore we only need to calculate the weight correction for weights connected to y_2 and y_5 .

$$\begin{aligned}
\Delta w_{12} &= \phi y_2 (\lambda x_1 - w_{12}) \\
&= 0.02(1) \left(\frac{0.1}{0.02}(0) - 0 \right) = 0 \\
\Delta w_{22} &= \phi y_2 (\lambda x_2 - w_{22}) \\
&= 0.02(1) \left(\frac{0.1}{0.02}(1) - 1 \right) = 0.08 \\
\Delta w_{32} &= \phi y_2 (\lambda x_3 - w_{32}) \\
&= 0.02(1) \left(\frac{0.1}{0.02}(0) - 0 \right) = 0 \\
\Delta w_{42} &= \phi y_2 (\lambda x_4 - w_{42}) \\
&= 0.02(1) \left(\frac{0.1}{0.02}(0) - 0 \right) = 0 \\
\Delta w_{52} &= \phi y_2 (\lambda x_5 - w_{52}) \\
&= 0.02(1) \left(\frac{0.1}{0.02}(1) - 0 \right) = 0.1
\end{aligned}$$

Repeat for the weights connected to y_5 .

$$\begin{aligned}
\Delta w_{15} &= \phi y_5 (\lambda x_1 - w_{15}) \\
&= 0.02(1) \left(\frac{0.1}{0.02}(0) - 0 \right) = 0 \\
\Delta w_{25} &= \phi y_5 (\lambda x_2 - w_{25}) \\
&= 0.02(1) \left(\frac{0.1}{0.02}(1) - 0 \right) = 0.1 \\
\Delta w_{35} &= \phi y_5 (\lambda x_3 - w_{35}) \\
&= 0.02(1) \left(\frac{0.1}{0.02}(0) - 0 \right) = 0 \\
\Delta w_{45} &= \phi y_5 (\lambda x_4 - w_{45}) \\
&= 0.02(1) \left(\frac{0.1}{0.02}(0) - 0 \right) = 0 \\
\Delta w_{55} &= \phi y_5 (\lambda x_5 - w_{55}) \\
&= 0.02(1) \left(\frac{0.1}{0.02}(1) - 1 \right) = 0.08
\end{aligned}$$

These calculations can also be presented in matrix form.

$$\begin{aligned}
\Delta w_{ij} &= \phi y_j (\lambda x_i - w_{ij}) \\
&= \alpha y_j x_i - \phi y_j w_{ij} \\
&= \alpha X Y^T - \phi W \text{diag}(Y)
\end{aligned}$$

$\text{diag}(Y)$ is the diagonal matrix of Y .

$$\begin{aligned}
\Delta w_{ij} &= \alpha \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \begin{bmatrix} y_1 & y_2 & y_3 & y_4 & y_5 \end{bmatrix} \\
&\quad - \phi \begin{bmatrix} w_{11} & w_{12} & w_{13} & w_{14} & w_{15} \\ w_{21} & w_{22} & w_{23} & w_{24} & w_{25} \\ w_{31} & w_{32} & w_{33} & w_{34} & w_{35} \\ w_{41} & w_{42} & w_{43} & w_{44} & w_{45} \\ w_{51} & w_{52} & w_{53} & w_{54} & w_{55} \end{bmatrix} \begin{bmatrix} y_1 & 0 & 0 & 0 & 0 \\ 0 & y_2 & 0 & 0 & 0 \\ 0 & 0 & y_3 & 0 & 0 \\ 0 & 0 & 0 & y_4 & 0 \\ 0 & 0 & 0 & 0 & y_5 \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
&= \alpha \begin{bmatrix} x_1y_1 & x_1y_2 & x_1y_3 & x_1y_4 & x_1y_5 \\ x_2y_1 & x_2y_2 & x_2y_3 & x_2y_4 & x_2y_5 \\ x_3y_1 & x_3y_2 & x_3y_3 & x_3y_4 & x_3y_5 \\ x_4y_1 & x_4y_2 & x_4y_3 & x_4y_4 & x_4y_5 \\ x_5y_1 & x_5y_2 & x_5y_3 & x_5y_4 & x_5y_5 \end{bmatrix} \\
&\quad - \phi \begin{bmatrix} w_{11}y_1 & w_{12}y_2 & w_{13}y_3 & w_{14}y_4 & w_{15}y_5 \\ w_{21}y_1 & w_{22}y_2 & w_{23}y_3 & w_{24}y_4 & w_{25}y_5 \\ w_{31}y_1 & w_{32}y_2 & w_{33}y_3 & w_{34}y_4 & w_{35}y_5 \\ w_{41}y_1 & w_{42}y_2 & w_{43}y_3 & w_{44}y_4 & w_{45}y_5 \\ w_{51}y_1 & w_{52}y_2 & w_{53}y_3 & w_{54}y_4 & w_{55}y_5 \end{bmatrix} \\
&= 0.1 \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} - 0.02 \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0.1 & 0 & 0 & 0.1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0.1 & 0 & 0 & 0.1 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0.02 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.02 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0.08 & 0 & 0 & 0.1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0.1 & 0 & 0 & 0.08 \end{bmatrix}
\end{aligned}$$

The updated weights would then be

$$\begin{aligned}
w_{ij} &= w_{ij} + \Delta w_{ij} \\
&= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0.08 & 0 & 0 & 0.1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0.1 & 0 & 0 & 0.08 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 + 0.08 & 0 & 0 & 0 + 0.1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 + 0.1 & 0 & 0 & 1 + 0.08 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1.08 & 0 & 0 & 0.1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0.1 & 0 & 0 & 1.08 \end{bmatrix}
\end{aligned}$$

Then the training process continues with Y_3 , Y_4 , and Y_5 , and repeat again with all the training data until the weights converge.

CSC3034 Current Course Update

07 hebbian calculation

Synchronized from the current CSC3034 course site on 14 August 2026.

- Current Lab 6 trains and evaluates a scikit-learn MLP on the Wine dataset.
- Current Lab 7 visualizes neural-network hyperplanes on standardized Iris features.
- Current Lab 8 adds Keras CNN/LSTM examples and Isaac Sim vision demonstrations.

Current materials author: Dr Tang Tiong Yew

See the synchronized lab notes and runnable examples for full instructions.